

★ WEEK 11 · LECTURE

Building Portfolios That Don't Break

Diversification, HRP, NCO — using the clean covariance from Week 9

Expanded edition · 38 slides · with three hands-on labs

Now we put money to work. The classic optimizer is fragile; modern methods cluster stocks into families and share risk fairly — staying stable even when data is noisy.

Where We Are Going Today

Week 9 cleaned the covariance; Week 10 kept us honest. Now we build the portfolio

1

The diversification problem

Spread risk across baskets. How we measure portfolio risk, and why we don't bet on one stock.

2

Why the classic method breaks

Markowitz inverts a noisy covariance and produces wild, fragile weights.

3

HRP and NCO

Cluster stocks into a family tree and share risk fairly — no risky matrix inversion.

By the end you will price the free lunch, watch Markowitz flip on a 0.5pp forecast, and run HRP by hand.

1

PART 1 · DIVERSIFICATION

The Diversification Problem

Spreading risk across baskets

Don't Put All Your Eggs in One Basket

The oldest rule in investing, made precise: spreading across stocks that don't all move together smooths the ride.

Portfolio Building = Spreading Eggs Across Baskets

All eggs in one basket



Put everything in one stock.

If it soars, you win big.

If it crashes, you lose everything.

A wild, dangerous ride.

Spread across many baskets



Split across many stocks.

One crash barely dents you.

The bumps cancel each other out.

A smoother, safer ride — same idea, less risk.

How We Measure a Portfolio's Risk

One number for the whole basket's bumpiness

$$\sigma_p^2 = w^\top \Sigma w \quad (\text{smaller} = \text{safer ride})$$

Reading it

- w = how much money sits in each stock (the weights).
- Sigma = the covariance: how the stocks move together.
- The result is the portfolio's overall risk (variance).

The key insight

- Risk is NOT just each stock's own wobble — it depends on how they move TOGETHER.
- Stocks that zig while others zag cancel out risk.
- So the covariance from Week 9 is the heart of it all.

The Free Lunch, Priced by Hand

Two stocks, 50/50: $\sigma_1 = 20\%$, $\sigma_2 = 10\%$ — only the correlation changes (full version: problem set)

Correlation ρ	$\sigma p^2 = 100 + 25 + 100\rho$ ($\%^2$)	Portfolio vol σp	Reading
+1 (clones)	225	15.0%	the plain average — no lunch at all
0 (strangers)	125	$\approx 11.2\%$	risk falls below the average — free lunch
-1 (opposites)	25	5.0%	umbrella + sunscreen — risk nearly vanishes

Same two stocks, same weights, same returns — vol from 15% down to 5%. Correlation IS the lunch menu.

Diversification Is About Correlation

Two stocks that move oppositely are a gift

If umbrella sales and sunscreen sales both sit in your portfolio, one always does well. Combining things that don't move together lowers risk WITHOUT lowering return.

The free lunch

- Diversification is the closest thing to a free lunch in finance.
- Lower risk for the same expected return.
- But it depends entirely on getting the correlations right.

The catch

- Those correlations are mostly noise (Week 9).
- Get them wrong and 'diversification' is an illusion.
- This is exactly where the classic method trips up.

Respect the Humble 1/N

Equal weight is embarrassingly hard to beat — every clever method must answer to it

Why 1/N is so strong

- It estimates NOTHING — so estimation error can't hurt it.
- It never concentrates: no single mistake can dominate.
- Out of sample, it routinely embarrasses optimized portfolios.

When cleverness earns its keep

- When risks genuinely differ: a calm bond next to a wild crypto.
- When structure is real: clusters that truly move together (W9).
- The bar for any method today: beat 1/N AFTER costs, out of sample.

1/N is the null hypothesis of portfolio construction — today's methods must reject it honestly.

2

PART 2 · THE FRAGILE CLASSIC

Why the Classic Method Breaks

Markowitz and the fragile inverse

Markowitz: Elegant, but Fragile

The 1952 idea that won a Nobel — and a hidden landmine

$$w^* \propto \Sigma^{-1} \mu \quad (\text{needs the inverse of } \Sigma)$$

Reading it

- It computes the 'optimal' weights w from returns and risk.
- μ = expected returns; Σ = the covariance.
- The trouble is that little -1 : it must INVERT the covariance.

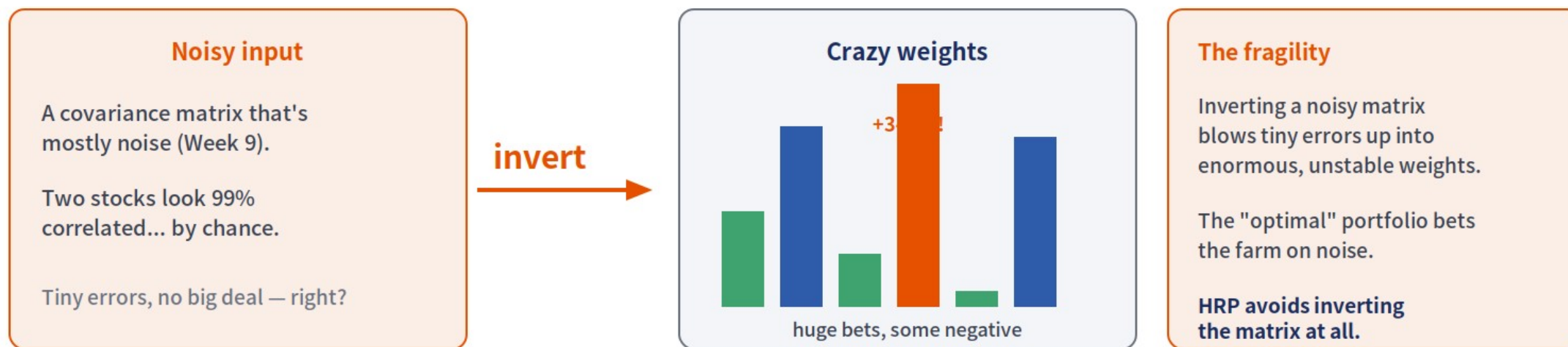
Why the inverse is dangerous

- Inverting amplifies the noise in Σ .
- Tiny estimation errors become giant weight errors.
- The 'optimal' portfolio bets the farm on noise — beautiful in theory, fragile in practice.

Noisy In, Crazy Out

Inverting a noisy matrix blows small errors into huge bets — why an 'optimal' portfolio often loses to an equal split.

Why the Classic Method (Markowitz) Breaks



Watch Markowitz Flip on a 0.5pp Forecast

Two near-twin stocks (σ equal, $\rho = 0.95$): $w \propto (\mu_1 - \rho\mu_2, \mu_2 - \rho\mu_1)$ (full version: problem set)

Forecast μ	Unnormalized $(\mu_1 - \rho\mu_2, \mu_2 - \rho\mu_1)$	Weights	Position in stock 2
(10%, 9.0%)	(1.45, -0.50)	(152.6%, -52.6%)	a -53% SHORT
(10%, 9.5%)	(0.975, 0)	(100%, 0%)	flat — the short vanished

A 0.5pp tweak to one forecast swung a position by 53 points. High correlation + that little -1 = a noise amplifier.

A Sturdier Building Block

Inverse-variance weighting: calmer stocks get more money — no inversion of the matrix

$$w_i = \frac{1/\sigma_i^2}{\sum_j 1/\sigma_j^2}$$

Reading it

- sigma squared = each stock's own riskiness.
- Calmer stock (small risk) → bigger weight.
- No matrix to invert, so nothing blows up.

Why it matters

- Simple, stable, hard to break.
- It's the basic weighting HRP uses inside each group.
- Robustness beats theoretical perfection on noisy data.

The Same Test, Given to IVP

Robustness has a definition: the answer barely moves when the inputs wobble

Question	Markowitz ($\rho = 0.95$)	Inverse-variance (IVP)
Weights with $\mu = (10\%, 9.0\%)$	(152.6%, -52.6%)	(50%, 50%) — μ is not even an input
Weights with $\mu = (10\%, 9.5\%)$	(100%, 0%) — a 53pp swing	(50%, 50%) — unmoved
With $\sigma^2 = (4, 1)$ instead	still forecast-hostage	(20%, 80%) — calm stock gets more

IVP ignores information (μ , correlations) and gains stability. HRP's insight: add the correlation back CAREFULLY, through a tree.

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PART 3 · HRP & NCO

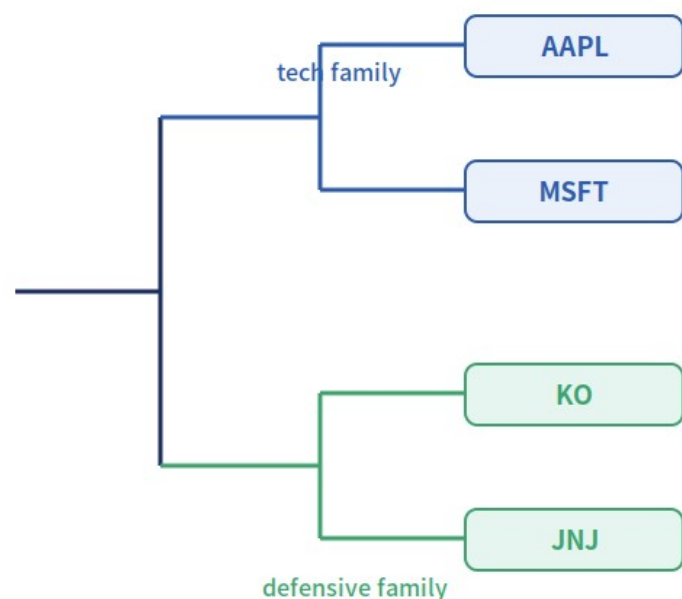
HRP & NCO

Cluster into families, share risk fairly

HRP — Hierarchical Risk Parity

Group stocks into a family tree, then split risk fairly down the branches. HRP never inverts the covariance.

HRP — Group Stocks into a Family Tree, Share Risk Fairly



Three simple steps — no matrix inverting

1. CLUSTER: group stocks that move alike into a tree (tech with tech, defensives with defensives).
2. ORDER: lay similar stocks next to each other.
3. SPLIT RISK: walk down the tree, dividing risk fairly between the two sides at each fork.

Result: sensible, stable weights — even with noisy data.

Why HRP Stays Stable

It respects the structure instead of fighting the noise

What Markowitz does

- Treats all stocks as one big tangled blob.
- Inverts the whole noisy matrix at once.
- One bad correlation poisons every weight — extreme, unstable bets.

What HRP does

- First groups similar stocks into a tree.
- Splits risk fairly at each fork, top down.
- A noisy correlation only affects its local branch — gentle, stable weights.

Marcos López de Prado introduced HRP in 2016; it often beats Markowitz out-of-sample.

HRP by Hand — Four Stocks, One Tree

Two defensives (corr 0) and two tech twins (corr 0.8), all with variance 1 (full version: problem set)

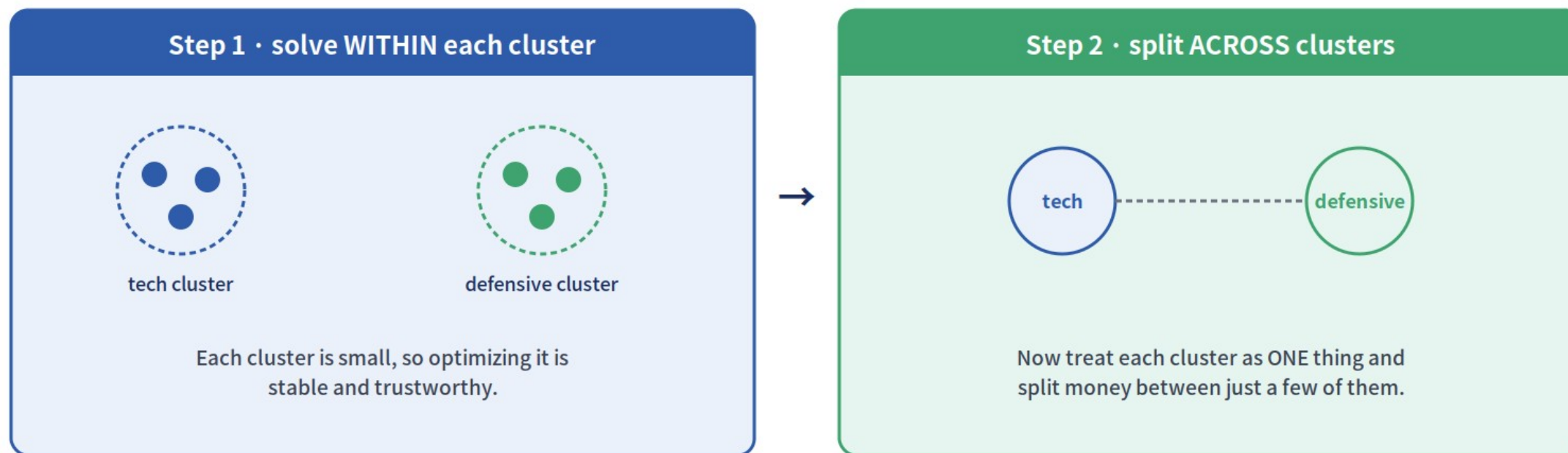
Step	Defensive cluster {KO, PEP}	Tech cluster {AAPL, AMZN}
Within: inverse-variance	50 / 50	50 / 50
Cluster variance $w'\Sigma w$	$0.25 + 0.25 = 0.5$	$0.25 + 0.25 + 2(0.25)(0.8) = 0.9$
Across: $\propto 1/\text{cluster var}$	$(1/0.5) \rightarrow 64.3\%$	$(1/0.9) \rightarrow 35.7\%$
Final weights	KO 32.1% · PEP 32.1%	AAPL 17.9% · AMZN 17.9%

Naive 1/N and plain IVP both say 25% each. HRP notices the tech twins are really ONE bet — and shifts to the true diversifiers.

NCO — Nested Clustered Optimization

Solve small problems inside clusters, then combine them. Breaking one giant optimization into small ones keeps every step stable.

NCO — Split One Big Problem into Small, Safe Ones



NCO in One Breath

Divide and conquer, the portfolio way

A giant optimization over 500 stocks is hopelessly noisy. But 50 small optimizations of 10 stocks each are stable — then you combine the 50 group results.

The two stages

- Within each cluster: optimize a small, stable group.
- Across clusters: treat each group as one asset, then split.
- Each step is small enough to trust.

HRP vs NCO

- HRP: split risk down a tree (no optimization inside).
- NCO: actually optimize inside each small cluster.
- Both use clustering to tame the noise.

Four Allocators on One Report Card

What each method uses, risks, and promises

	1/N	IVP	Markowitz	HRP / NCO
Inverts the covariance?	no	no	YES — the landmine	no / only small blocks
Uses forecasts μ ?	no	no	yes — forecast-hostage	no / optional
Sees correlations?	no	no	all at once, raw	through the tree, tamed
Stability on noisy data	excellent	excellent	fragile	strong
The catch	ignores everything	ignores correlation	noise amplifier	needs honest clustering

The pattern of the whole course: use MORE information only as carefully as you can estimate it.

Stability Is Not Aesthetic — It Is Money

Two invoices that fragile weights quietly send you

Invoice 1: turnover

- Fragile weights lurch every time estimates update.
- Every lurch is a trade; every trade pays spread, impact, tax.
- A 'better' portfolio can lose its edge in trading costs alone.

Invoice 2: whipsaw & trust

- Extreme bets force panicked rebalances at the worst moments.
- Committees abandon strategies they cannot explain (IC reality).
- Stable, explainable weights survive both markets and meetings.

HRP's calm weights are a feature you can invoice: lower costs, fewer panics, longer survival.

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PART 4 · PUTTING IT TOGETHER

Putting It Together

From clean covariance to a real portfolio

The Whole Pipeline, End to End

Every recent week clicks into place here

1

Estimate & clean the covariance (Week 9)

Build the correlation matrix, then denoise it — flatten the random noise, keep the real signal.

2

Cluster the stocks into families (Week 9)

Use the clustering idea to find natural groups that move together.

3

Build robust weights with HRP or NCO

Share risk fairly down the tree — no fragile matrix inversion, stable even with noisy data.

Clean inputs + a robust method = a portfolio that survives live trading, not just the backtest.

The Quant Production Line, Weeks 1–11

Today is where the assembly line finally ships a product

Station	Week	What it contributes to today's portfolio
Raw material	W2	honest bars, survivorship-aware universes
Labeling	W3	triple-barrier outcomes worth predicting
Signal models	W4–W8	forecasts and anomaly screens (used with suspicion)
Clean risk inputs	W9	denoised covariance + the cluster tree
Quality control	W10	purged CV + DSR before anything ships
Assembly	W11	HRP / NCO turn it all into weights

No station is optional — most blow-ups trace to skipping quality control, not to a bad formula.

Where This Lives in the Real World

Robust portfolios for institutions

Why institutions love HRP/NCO

- They manage hundreds of assets; stable weights mean fewer panicked rebalances.
- Lower trading costs and turnover.
- Easier to explain than a black-box optimizer.

The honest caveat

- No method is magic; robustness reduces blow-ups, it doesn't promise profit.
- Always pair with the honest testing from Week 10.
- Clean data and discipline matter more than the formula.

The goal isn't the 'best' backtested portfolio — it's one that behaves sensibly when reality hits.

Today's Story in Four Numbers

Fragility and robustness, measured on this week's own examples

53pp

Markowitz position
swing caused by a
0.5pp forecast change

5%

Two-stock portfolio
vol at $\rho = -1$ (from
15% at $\rho = +1$)

64.3%

HRP's allocation to
the truly-diversifying
cluster

2016

HRP's debut —
portfolio theory
learned to stop
inverting

One thread: the less your answer moves when noise wiggles the inputs, the more it is worth.

Which Allocator? — A 60-Second Drill

Cover the right column, answer out loud, then check yourself

Scenario	Which tool / diagnosis?
No estimates you trust at all	1/N — estimation error can't touch it
A calm bond fund next to a wild crypto	IVP — calmer sleeve gets more money
Optimizer outputs +152% / -53% on twins	Markowitz fragility — the inverse amplified noise
Ten stocks, two tight clusters, noisy corr	HRP — split risk down the tree, no inversion
500 names, real structure, want optimization	NCO — optimize small clusters, then combine

If you got 4 of 5, the allocator's instinct of Week 11 is yours.

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PART 5 · HANDS ON

Your Turn — Hands On

Build HRP weights and race the allocators

Getting This Week's Data

Companion notebook — the multi-stock panel (§ 3b)

From the companion notebook (§ 3b)

- Run Setup (§ 0), then the § 3(b) panel loader.
- Daily returns for ~10 stocks in a tidy table `rets`.
- Same panel as Weeks 4 and 9 — connect the ideas.

What we'll do

- Lab 1: build HRP weights via clustering + inverse-variance.
- Lab 2: compare risk — equal vs inverse-var vs HRP.
- All in `scipy` + `numpy`, a few lines each.

[CODE] FML_Data_Setup_Guide.ipynb · § 3 (b) panel loader

```
import yfinance as yf
tickers = ['AAPL', 'MSFT', 'GOOGL', 'AMZN', 'META', 'KO', 'PEP', 'XOM', 'JPM', 'JNJ']
prices = yf.download(tickers, start='2020-01-01', end='2024-12-31',
                    auto_adjust=True, progress=False)['Close'].dropna()
rets = prices.pct_change().dropna() # rows = days, cols = stocks
```

Hands-On Lab 1 — Build HRP Weights

Cluster the stocks, then share risk down the tree

[CODE] colab_lab1.py · data from setup guide § 3b

```
import numpy as np
from scipy.cluster.hierarchy import linkage, leaves_list

corr = rets.corr().values
dist = np.sqrt(0.5 * (1 - corr)) # corr -> distance
order = leaves_list(linkage(dist, 'single')) # tree order

# simple HRP-style weight: calmer stock -> more money
var = rets.var().values[order]
w = (1 / var) / np.sum(1 / var)
for name, wi in zip(rets.columns[order], w):
    print(f'{name:6s} {wi:6.1%}')
```

Your task (step by step)

- Load the panel rets (setup guide § 3b).
- Cluster stocks by correlation (scipy linkage).
- Order them by the tree.
- Weight calmer stocks more (inverse-variance).
- Q: are the weights gentle and sensible?

Hands-On Lab 2 — Compare Three Portfolios

Equal weight vs inverse-variance vs HRP-style

[CODE] colab_lab2.py · data from setup guide § 3b

```
import numpy as np
cov = rets.cov().values
n = rets.shape[1]

def risk(w):
    return np.sqrt(w @ cov @ w)    # portfolio volatility

equal = np.ones(n) / n
var = np.diag(cov)
invvar = (1/var) / np.sum(1/var)
print('equal weight risk:', round(risk(equal), 5))
print('inverse-var risk:', round(risk(invvar), 5))
# HRP weights from Lab 1 land in between, but stay stable
```

Your task (step by step)

- Build three sets of weights.
- Compute each portfolio's risk (volatility).
- Compare them side by side.
- See which gives the smoothest ride.
- Q: does smarter weighting lower the risk?

Hands-On Lab 3 — Claude as Your Coding Copilot

AI is allowed in this course — it explains and extends; you understand and verify

[INPUT] Prompt · Claude (paste after your Lab 1 + 2 code)

You are my machine-learning tutor.

- 1) Explain my Lab 1 and Lab 2 code above, one plain-English sentence per line.
- 2) Implement FULL HRP: correlation-distance linkage, quasi-diagonalization, recursive bisection with cluster variance; print the weights.
- 3) Race four allocators - 1/N, IVP, HRP, min-variance on the RAW covariance - and a fifth: min-variance on the Week 9 DENOISED covariance. Table: in-sample vol, max weight, min weight.
- 4) Stability test: re-estimate on 20 bootstrap resamples; report each allocator's average weight turnover across resamples.
- 5) In 3 sentences: which allocator wins on stability, and what denoising did for min-variance.

House rules for AI use

- Allowed: explain, debug, extend code you wrote.
- Required: run it, read it, defend every line.
- Forbidden: submitting code you cannot explain.
- Tip: ask 'why', not just 'do'.

Comprehension Check

Try each question first — the answer is revealed below

Q1 Why can the classic Markowitz method produce crazy weights?

- A** It uses too few stocks
- B** Inverting a noisy covariance blows up tiny errors ✓
- C** It ignores returns
- D** It needs labels

Answer: Inverting a noise-filled covariance amplifies small errors into huge, unstable bets.

Q2 What is HRP's key trick for staying stable?

- A** It inverts the matrix twice
- B** It clusters stocks into a tree and never inverts the matrix ✓
- C** It uses only one stock
- D** It shuffles the data

Answer: HRP groups stocks into a family tree and splits risk down it — avoiding the fragile matrix inversion.

Comprehension Check — Round Two

Two more — these cover the new slides

Q3 A 0.5pp forecast change flipped a −53% short to flat. The culprit:

- A Too little data
- B Σ^{-1} amplifying noise between near-twins ($\rho = 0.95$) ✓**
- C The risk-free rate
- D Too many clusters

Answer: Near-identical assets make the inverse explosive: tiny μ differences become giant opposing bets.

Q4 HRP gave the tech twins (corr 0.8) LESS money than the defensives. Why?

- A Tech is riskier by law
- B Their cluster variance is higher — they are really one bet ✓**
- C Alphabetical order
- D It maximizes returns

Answer: Correlated twins diversify each other poorly: cluster variance 0.9 vs 0.5 → the split leans to the true diversifiers.

Week 11 on One Page

Photograph this slide — it is the whole lecture

Concept	One line to remember
Portfolio risk	$w'\Sigma w$ — it lives in the co-movements, not the individual wobbles.
Free lunch	same stocks, ρ from +1 to -1: vol 15% $\rightarrow$ 5%. Correlation is the menu.
Markowitz	$w \propto \Sigma^{-1}\mu$ — elegant, but the inverse amplifies noise into extreme bets.
IVP	$w \propto 1/\sigma^2$ — ignores μ and ρ , gains stability; HRP's building block.
HRP	cluster $\rightarrow$ within-IVP $\rightarrow$ split across by $1/(\text{cluster variance})$. Never inverts.
NCO	optimize inside small clusters, then across — divide and conquer.

Next stop: teaching machines to read — NLP, LLMs, and a finance chatbot.

Three Ideas, One Picture Each

Week 9 cleaned the inputs; Week 11 builds robustly on them. Clean data + robust method wins

1

Diversification = spread the eggs

Portfolio risk depends on how stocks move together. Combine things that don't move alike to smooth the ride.

2

Markowitz is fragile

It inverts a noisy covariance and produces wild, unstable weights — often worse than a simple equal split.

3

HRP & NCO cluster, then share risk

Group stocks into a family tree and divide risk fairly — robust even when the data is noisy.

Keep the pictures: eggs in baskets, a noise-amplifying inverse — and a family tree sharing risk.

Week 11 — Three Things to Remember

Before next class: finish Labs 1–3 and compare the HRP weights to a naive equal split

1

Risk is about moving together

A portfolio's risk is $w^T \Sigma w$. Diversification works when holdings don't all move alike.

2

Don't invert a noisy matrix

Markowitz's optimal weights need the covariance inverse, which amplifies noise into fragile, extreme bets.

3

Cluster, then allocate (HRP/NCO)

Group stocks into families and share risk fairly. Robust, stable, and built on Week 9's clean covariance.

Next week — NLP, LLMs, RAG: teaching machines to read, and building a finance chatbot.